ETHICAL USE OF AI IN ACADEMIC RESEARCH

5 TIPS FOR USING AI IN RESEARCH

01 DON'T ASK AI TO CHOOSE YOUR TOPIC

Generate brainstorming ideas, but make the final choice yourself based on your interests and available data.

PERPLEXITY AI

02 UNDERSTAND PAPERS FASTER

Use AI to quickly summarize complex data, methodology, and core findings, saving significant reading time.

ELICIT · R DISCOVERY · CONNECTED PAPERS

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03 CRITICALLY EVALUATE A PAPER

Analyze key arguments, strengths, and limitations with AI help to form a critique.

SCISPACE

04 ASK SPECIFIC QUESTIONS

Query AI models for precise insights on particular paragraphs, data points, or citations.

NOTEBOOKLM

05 NEVER ASSUME AI IS CORRECT

Fact-check and verify every AI generation against the source. AI can produce incorrect summaries and data.

5 TIPS FOR USING AI IN RESEARCH

1. DON'T ASK AI TO CHOOSE YOUR RESEARCH TOPIC FOR YOU.

Start with your own interests, background, and research problem. Give AI the context it needs to help you refine your idea.

Tool: Perplexity AI

Use: To explore a topic, identify relevant debates, and discover possible research directions. Verify sources before relying on them.

Prompt: "I want to research [domain]. My academic background is [background], and I am particularly interested in [area]. Suggest possible research topics and explain how each could be narrowed down."

2. USE AI TO UNDERSTAND RESEARCH PAPERS FASTER.

Upload a paper and ask AI to summarize its central argument, research question, methodology, and findings. This can help you understand the paper's overall context before reading it closely.

Tool: Elicit – Find relevant studies and extract key research details.

Tool: R Discovery – Discover relevant academic literature and get article summaries.

Tool: Connected Papers – Map related studies and identify connections between papers.

3. USE AI TO CRITICALLY EVALUATE A PAPER.

Ask AI to identify the paper's strengths, limitations, assumptions, methodological gaps, and unanswered questions. Then compare its assessment with the paper's actual arguments and evidence.

Tool: SciSpace

Use: Explore paper content and ask questions about its methodology, evidence, and limitations.

Prompt: "Critically assess this paper's methodology, evidence, limitations and contribution to the existing literature."

Key Takeaway: Treat AI's critique as a starting point for your own evaluation.

4. ASK SPECIFIC QUESTIONS ABOUT THE PAPER.

Confused by a theoretical framework, a statistical result, or a methodological choice? Ask targeted questions about the relevant section rather than requesting a generic summary.

Tool: NotebookLM

Use: Upload a paper and ask specific questions about the relevant section, such as the theoretical framework, methodology, or results.

Prompt: "Explain why the authors have used this theoretical framework and how it supports their central argument."

5. NEVER ASSUME AI IS ALWAYS CORRECT.

AI can misinterpret arguments, produce incorrect citations, and present unsupported claims confidently. Verify facts against original sources, check references, and independently assess its answers.

Golden Rule: Always return to the original source before using information in your research.